DIGITAL SAT & EST MATH COURSES

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EST1 October 2020

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Algebra, functions, data analysis, probability and geometry from the official EST I paper. Work through all 58 questions under timed conditions, then check your answers with the answer key and the step-by-step solutions at the end.

58	MCQ	90 min
QUESTIONS	MULTIPLE CHOICE	TIME LIMIT

NAME _____ DATE _____ SCORE _____ /58

How to get the most from this practice test

- **Read** every question slowly and underline the key words.
- **Draw** a quick sketch for every geometry question before you solve it.
- **Solve** step by step and show your work.
- **Check** your answer by substituting it back before you choose.

YOUR MATH COACH

RF

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Digital SAT & EST Math expert with **23 years of experience**, helping students master math and reach **700+**.

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Practice Questions

58 QUESTIONS

1

1. Isolate **r**.

A. $r = \frac{4\pi\varepsilon_0 F}{q_1 q_2}$

B. $r = \sqrt{\frac{q_1 q_2 F}{4\pi\varepsilon_0}}$

C. $r = \frac{1}{2}\sqrt{\frac{q_1 q_2}{\pi\varepsilon_0 F}}$

D. $r = (\frac{q_1 q_2}{4\pi\varepsilon_0})^2$

2

2. If the charges are to be separated by a distance of **3r**, an electrostatic force F_1 is created. What is the relationship between F_1 and F?

A. $F_1 = 3F$
B. $F_1 = F/3$
C. $F_1 = 9F$
D. $F_1 = F/9$

3

If $f(x) = x^2 - 5x - 6$ and $g(x) = -3 - x$, what is the value of $f(g(-1))$?

(A) 13
(B) 8
(C) 0
(D) 30

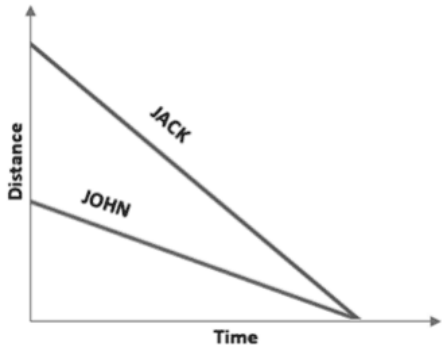
4

In the xy-plane, the equation of the line (d) passing through $A(1, 3)$ and perpendicular to line $3x - 2y = 6$ is:

(A) $y = \frac{3}{2}x - \frac{7}{2}$
(B) $y = -\frac{3}{2}x + \frac{11}{2}$
(C) $y = -\frac{2}{3}x + 3$
(D) $y = -\frac{2}{3}x + \frac{11}{3}$

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5 Jack and John are meeting at a restaurant. The scheme above represents the drives from their offices to the restaurant. Which of the following statements is true?



A. It took Jack longer to arrive to the pub since his office is farther away.
B. It took John longer to arrive to the pub since his office is farther away.
C. John drove to the pub at a faster speed than Jack.
D. Jack drove to the pub at a faster speed than John.

6 Which value of x makes the equation $\frac{7}{5}(3x - 2) = 14$ true?

A. 2
B. 3
C. 4
D. 5

7 A scatter plot shows a strong positive correlation between two variables: A graphed on the x-axis and B graphed on the y-axis. Which of the following statements is justified?

A. The rise in A caused the rise in B.
B. The rise in B caused the rise in A.
C. A&B are correlated, but causation is unknown.
D. A&B rise proportionally.

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8 A game consists of throwing a dart into a target divided into 2 sections: The inner section rewards a player with 5 points, while the outer section gets him 3 points only. Kurt throws 6 darts and hits the inner section x times. What is Kurt's total score in terms of x?



(A) $8x + 24$
(B) $2x + 18$
(C) $-2x + 18$
(D) $6x + 4$

9 **Questions 9 and 10 refer to the following information.**

We survey 125 employees at random from each of two companies labeled **A** and **B**, and separated into groups based on how many siblings do they have. The results are shown in the table below.

Number of siblings	Company A	Company B
0	15	25
1	45	40
2	30	25
3	25	15
4	10	20

9. What is the median number of siblings of the sample of employees in company B?

A. 1
B. 2
C. 3
D. 4

10 **10.** Which measure of center is the same in both companies?

A. Range
B. Mean
C. Median
D. Mode

11 If $f(x) = x^2 - 5x - 6$ and $g(x) = -3 - x$, what is the value of $f(g(-1))$?

(A) 13
(B) 8
(C) 0
(D) 30

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12 $y = \frac{x^3-3x^2+2x-1}{x-3}$

Which of the following expressions is equivalent to y?

A $y = x^2 + 2 + \frac{5}{x-3}$

B $y = x^2 - 2x + 2 + \frac{3}{x-3}$

C $y = x^2 - 3 - \frac{8}{x-3}$

D $y = x^2 + 2$

13 The graph (C) of the function $f(x) = 2(x+2)(x-6)$ is a parabola. If the line $x = k$ is the axis of symmetry of the parabola, what is the value of k?

A 1

B 2

C 3

D 4

14 **14.** If $\frac{1}{x-y} = \frac{3}{5y}$, which of the following proportions is equivalent?

A. $\frac{x}{y} = \frac{3}{8}$

B. $\frac{x}{y} = \frac{8}{3}$

C. $\frac{x}{y} = \frac{8}{15}$

D. $\frac{x}{y} = \frac{15}{8}$

15 In 2017, the number of people who had access to the internet in a country was 3.2 million. If this number grows at a rate of 18% each year, which expression best describes the number of people y, in million, having access to the internet x years after 2017?

A $y = 0.18x + 32$

B $y = 1.18x + 32$

C $y = 3.2(0.18)^x$

D $y = 3.2(1.18)^x$

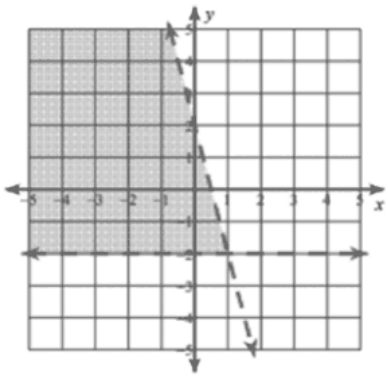
16 If $f(x) = x^2 - 5x - 6$ and $g(x) = -3 - x$, what is the value of $f(g(-1))$?

X	2	5	-3
Y	-1	-7	?

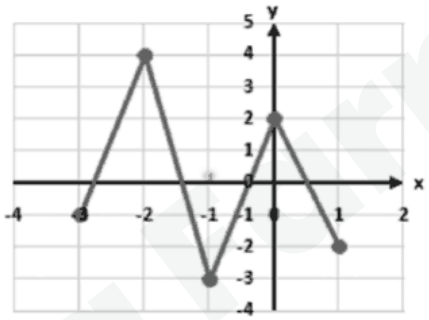
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17 In the coordinate plane, the shaded region is the solution of the system $\begin{cases} 4x + y < 2 \\ y \geq -2 \end{cases}$

For $x = 0$, what is the highest integer value of y?



18 Based on the graph of the function $f(x)$, if the absolute maximum and minimum of the represented function $f(x)$ are (p, q) and (r, s) respectively, what is the value of $q - p + s - r$?



19 If $f(x) = x^2 - 5x - 6$ and $g(x) = -3 - x$, what is the value of $f(g(-1))$?

20 If $f(x) = x^2 - 5x - 6$ and $g(x) = -3 - x$, what is the value of $f(g(-1))$?

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21 The secret value, k, of a 4-digit pin code $abcd$ is obtained by subtracting the triple of b from c, and dividing the resulting expression by half of the sum of a and d. What is the secret value, k, in terms of a, b, c and d?

(A) $k = \frac{c-3b}{2a+2d}$
(B) $k = \frac{b-3c}{2a+2b}$
(C) $k = \frac{2c-6b}{a+d}$
(D) $k = \frac{6b-2c}{a+d}$

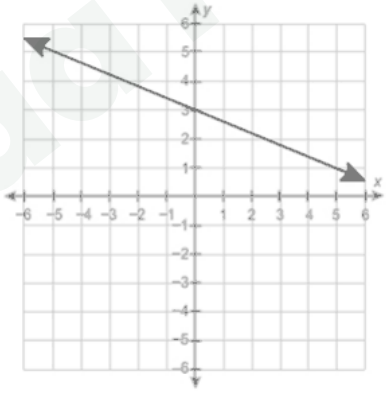
22 A library sells new and used books. If, out of the total of 474, there are twice as many new books as old ones. How many new books are there in the library?

(A) 316
(B) 158
(C) 352
(D) 238

23 What is the area of the sector defined by a central angle of $54°$ in the circle of equation $x^2 + y^2 - 8x + 12y - 12 = 0$?

(A) 15π
(B) 9.6π
(C) 5.4π
(D) 2.4π

24 The graph shown represents which of the following equations?



(A) $y = -\frac{2}{5}x - 3$
(B) $y = \frac{2}{5}x + 3$
(C) $y = -\frac{2}{5}x + 3$
(D) $y = -\frac{5}{2}x + 3$

25 If $f(x) = 5 - 2x$ and $g(x) = \frac{x^2}{4}$, Which of the following is not in the range of $f(g(x))$?

(A) -3
(B) 0
(C) 5
(D) 6

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26

6. A line, *having* a slope of $-\frac{2}{3}$ passes through the points A (2-k ; 5) and B (-2k ; -1). What is the value of k?

A. 11
B. 4
C. -4
D. -11

27

In which major can you find the highest percentage of students who prefer NYC?

	History	Math	Audit	IT	Total
Rome	65	35	35	15	**150**
Paris	75	65	10	25	**175**
Tokyo	40	115	50	20	**225**
NYC	70	60	30	40	**200**
Total	**250**	**275**	**125**	**100**	**750**

(A) History
(B) IT
(C) Audit
(D) Math

28

What fraction of people who prefer Rome or Paris come from Audit or Math majors?

(A) 140/325
(B) 17/65
(C) 29/65
(D) 4/13

29

The selling price of an apartment is directly proportional its area. If an apartment of 127 sqm is sold for $\$168,275$, what would be the price of a 156 sqm apartment?

(A) $237,900\$$
(B) $208,500\$$
(C) $136,993\$$
(D) $206,700\$$

30

In a hospital, the medical staff is composed of 34 doctors and 68 nurses. How many additional doctors should join the hospital in order for the ratio of doctors to total number of medical staff to become 3 to 7?

(A) 21
(B) 19
(C) 17
(D) 15

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31 $\sqrt{27x^3y^5 - 18x^2y^4}$

Which of the following is equivalent to the above expression? (x and $y > 0$)

(A) $3xy^2(\sqrt{3xy} - \sqrt{2})$
(B) $3xy^2\sqrt{3xy - 2}$
(C) $9xy^2\sqrt{3xy - 2}$
(D) $9xy^2(\sqrt{3xy} - \sqrt{2})$

32 If $f(x) = x^2 - 5x - 6$ and $g(x) = -3 - x$, what is the value of $f(g(-1))$?

(A) 13
(B) 8
(C) 0
(D) 30

33 Maria downloaded to her music library a total of 350 pop and rock songs. If the ratio of pop to rock song is 3 to 11. How many rock songs are there in Maria's library?

(A) 75
(B) 125
(C) 175
(D) 275

34 **14.** If $5 - \frac{3}{2}x \geq 3$, what is the highest value of $\frac{9}{8}x + 1$?

A. 2.5
B. 3.5
C. 4.5
D. 5.5

35 A craftsman is looking for two kinds of paint from a wholesaler. The first kind a is packaged in 10 kg jars, the second b in 25 kg jars. The 10 kg jar costs $\$45$ and the 25 kg one costs $\$120$. The load must not exceed 250 kg and the total sum must be at least $\$900$ in order to get a discount. Which system of inequalities verifies the given information?

(A) $\begin{cases} 10a + 25b \geq 250 \\ 45a + 120b \leq 900 \end{cases}$
(B) $\begin{cases} 10a + 25b \leq 250 \\ 45a + 120b \leq 900 \end{cases}$
(C) $\begin{cases} 10a + 25b \geq 250 \\ 45a + 120b \geq 900 \end{cases}$
(D) $\begin{cases} 10a + 25b \leq 250 \\ 45a + 120b \geq 900 \end{cases}$

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36 Brad got an average of 76 on his last three math tests. What grade should he get on the fourth test to obtain an average of 80?

(A) 88
(B) 90
(C) 92
(D) 94

37 From 2018 to 2019, the amount in Julias's bank account increased by 22.5% to $14325. To the nearest dollar, what was the initial amount in her bank account?

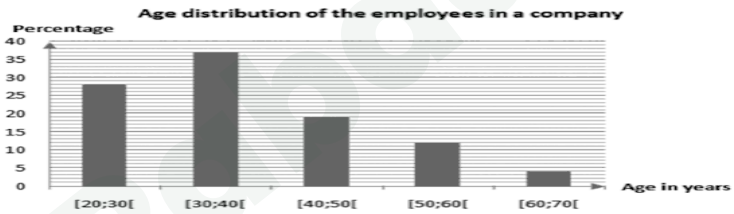
(A) 11694$
(B) 14010$
(C) 11102$
(D) 12775$

38 The ministry of finance conducted a survey to determine the average income rate in the industry of banking in a given country. So, a meeting was held in the presence of 150 bank managers to ask them about their annual income. Which conclusion can be drawn about the survey's reliability?

(A) The survey is reliable because it involves people from the banking industry.
(B) The survey is unreliable because the sample is too small.
(C) The survey is unreliable because the sample does not cover employees of different positions.
(D) The survey is reliable since it was made by the ministry of finance.

39 **Questions 19 and 20 refer to the following information.**

The bar graph below shows the age distribution of the employees in a multi-branched bank.



19. If 481 employees are aged between 30 and 40, what is the total number of employees in this company?

A. 1150
B. 1200
C. 1550
D. 1300

40 In which interval does the median age lie?

(A) [20;30[
(B) [30;40[
(C) [40;50[
(D) [50;60[

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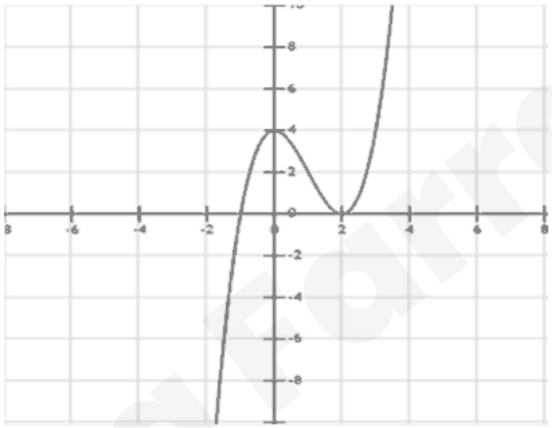
41

$$\begin{cases} \frac{2}{5}x - \frac{1}{3}y = 7 \\ -\frac{m}{10}x + \frac{5}{6}y = 3 \end{cases}$$

21. If the system of linear equations above admits no solutions, and m is an integer, what is the value of m?

A. -2
B. 10
C. 6
D. -10

42 The graph plotted above represents which of the following functions?



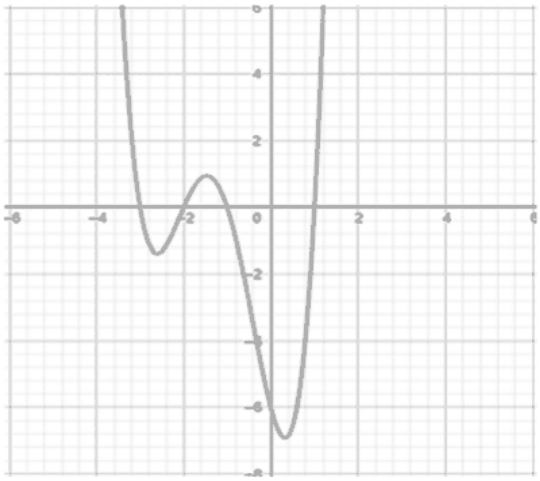
(A) $f(x) = (x+1)(x-2)^2$
(B) $f(x) = (x-1)(x+2)^2$
(C) $f(x) = (x+1)(x-2)$
(D) $f(x) = (x-2)(x+1)^2$

43 A secretary types on her computer keyboard at an average speed of 1.75 words per second. Her manager asked her to type a document containing 16 pages with an average of 525 words per page. How long will she be actively typing this document?

(A) 1 hour, 10 minutes
(B) 1 hour, 20 minutes
(C) 2 hours, 5 minutes
(D) 4 hours, 5 minutes

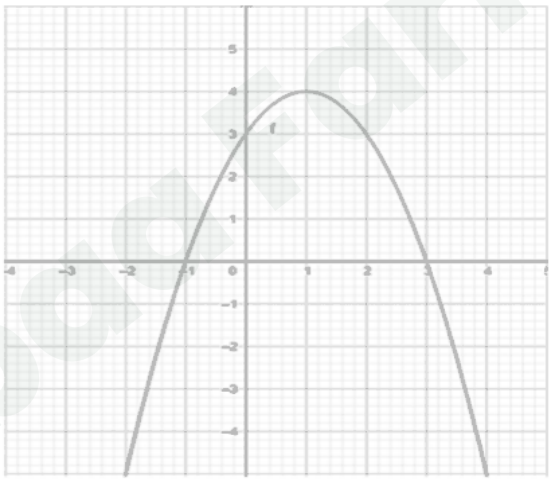
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44 A graph above depicts a function $f(x)$. How many solutions does the equation $f(x) = 0.5$ admit?



(A) 1
(B) 2
(C) 3
(D) 4

45 What is the equation of the function f graphed above?



(A) $f(x) = (x + 1)(x - 3)$
(B) $f(x) = (x - 1)(x + 3)$
(C) $f(x) = -(x + 1)(x - 3)$
(D) $f(x) = -(x - 1)(x + 3)$

46 If $f(x) = x^2 - 5x - 6$ and $g(x) = -3 - x$, what is the value of $f(g(-1))$?

(A) 13
(B) 8
(C) 0
(D) 30

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47 A liquid covering 12 cm of the height of a cylindrical glass of diameter 8cm is transferred into another glass shaped as a cone with a diameter of 12 cm. What height of the cone will the liquid cover?

(A) 8 cm
(B) 10 cm
(C) 16 cm
(D) 14 cm

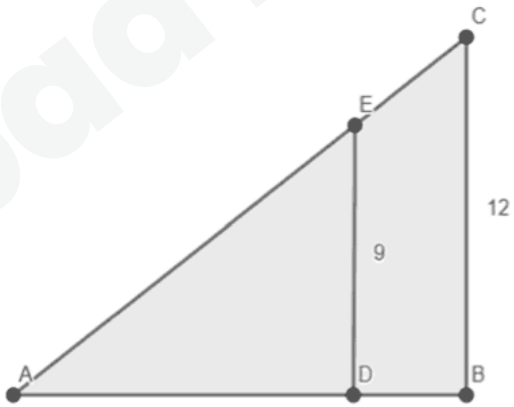
48 A gaming website charges its client by number of games downloaded g. If the total fees f of a gamer are given by the equation $f = 7g + 88$, what is the best interpretation of the term 88?

(A) The price of one downloaded game.
(B) The subscription fees of the website
(C) The average number of games downloaded per player
(D) The fees of downloading 7 games

49 A local supermarket offered a discount of 15% on its items after the government raised all the prices by 25%. Overall, by what percentage were the original prices raised in this supermarket in particular?

(A) 8.625%
(B) 7.250%
(C) 6.250%
(D) 5.625%

50 The right triangles ADE and ABC are similar. If $AD = \frac{4}{3}ED$, how long is DB?



(A) 3
(B) 4
(C) 5
(D) 6

51 If $f(x) = x^2 - 5x - 6$ and $g(x) = -3 - x$, what is the value of $f(g(-1))$?

52 A multiple choice test contains 50 questions. A correct answer is worth 3 points and an incorrect answer is worth -2 points. If a student receives 75 on the test, how many questions did he answer correctly?

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53 If $|2b - 1| \leq 3$, how many possible integer values of b are there?

54 We chose randomly a sample of 250 lawyers out of the 4900 registered in the syndicate, and we asked them about the most frequent case they defend. The results showed that 45 answered divorce, 125 answered theft and 80 answered murder. Presumably, what is the total number of lawyers registered in the syndicate who deal most frequently with murder?

55 The function g is defined by $g(x) = ax^2 - 2x - 5$ and $g(-1) = 1$. What is the value of $g(2)$?

56 What is the remainder of the division of $k(x) = 3x^3 + 8x^2 - 2x - 7$ by $x + 2$?

57 If $f(x) = x^2 - 5x - 6$ and $g(x) = -3 - x$, what is the value of $f(g(-1))$?

58 A right triangle has an area of $96\ \text{cm}^2$. If the shorter leg is $4\ \text{cm}$ less than the longer leg, what is the length of the hypotenuse?

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Answer Key

58 ANSWERS

Q	ANSWER	Q	ANSWER
1	C	30	C
2	D	31	B
3	B	32	D
4	D	33	D
5	D	34	A
6	C	35	D
7	C	36	C
8	B	37	A
9	A	38	C
10	D	39	D
11	B	40	B
12	A	41	B
13	B	42	A
14	B	43	B
15	D	44	D
16	9	45	C
17	1	46	A
18	4	47	C
19	8	48	B
20	4/5	49	C
21	C	50	B
22	A	51	25/8
23	B	52	35
24	C	53	4
25	D	54	1568
26	D	55	7

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Detailed Solutions

58 SOLUTIONS

1 **Answer: C**

See the answer key.

2 **Answer: D**

See the answer key.

3 **Answer: B** 8

First, evaluate $g(-1)$:
$g(-1) = -3 - (-1) = -3 + 1 = -2$
Next, substitute this result into $f(x)$:
$f(-2) = (-2)^2 - 5(-2) - 6 = 4 + 10 - 6 = 8$

4 **Answer: D** $y = -\frac{2}{3}x + \frac{11}{3}$

The given line is $3x - 2y = 6$. Rewrite it in slope-intercept form:
$-2y = -3x + 6 \implies y = \frac{3}{2}x - 3$.
The slope is $\frac{3}{2}$. The slope of a perpendicular line is the negative reciprocal, $-\frac{2}{3}$.
Use point-slope form with $A(1, 3)$:
$y - 3 = -\frac{2}{3}(x - 1)$
$y = -\frac{2}{3}x + \frac{2}{3} + 3 = -\frac{2}{3}x + \frac{11}{3}$

5 **Answer: D** Jack drove to the pub at a faster speed than John.

Based on the graph:
- The y-intercept represents the initial distance. Jack's initial distance is greater, so his office is farther away.
- The x-intercept represents the time of arrival. Both lines meet the x-axis at the same point, meaning they arrived at the same time. This eliminates A and B.
- Speed is Distance/Time (the magnitude of the slope). Since Jack traveled a greater distance in the same amount of time, he must have driven at a faster speed.

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6 **Answer: C** 4

$\frac{7}{5}(3x-2)=14$
Multiply both sides by $\frac{5}{7}$:
$3x-2=14\times\frac{5}{7}$
$3x-2=10$
$3x=12$
$x=4$
Wait, the correct option for $x=4$ is C, let me fix. It asks `Which value of x makes...`. so $x=4$, corresponding to option C. I will adjust the correctAnswer.

7 **Answer: C** A&B are correlated, but causation is unknown.

A fundamental principle in statistics is that correlation does not imply causation. Two variables can be strongly correlated without one causing the other. Therefore, A and B are correlated, but causation is unknown.

8 **Answer: B** $2x+18$

He threw 6 darts total. He hit the inner section x times, so he hit the outer section $6-x$ times.
Total score = $5x+3(6-x)$
Score = $5x+18-3x=2x+18$

9 **Answer: A**

See the answer key.

10 **Answer: D**

See the answer key.

11 **Answer: B** 8

First, evaluate $g(-1)$:
$g(-1)=-3-(-1)=-3+1=-2$
Next, substitute this result into $f(x)$:
$f(-2)=(-2)^2-5(-2)-6=4+10-6=8$

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12 **Answer: A** $y = x^2 + 2 + \frac{5}{x-3}$

Use polynomial long division or synthetic division to divide $x^3 - 3x^2 + 2x - 1$ by $x - 3$.
Dividing $x^3 - 3x^2$ by $x - 3$ gives x^2.
Subtracting $x^2(x - 3)$ leaves $2x - 1$.
Dividing $2x$ by x gives 2.
Subtracting $2(x - 3) = 2x - 6$ leaves a remainder of 5.
So the result is $x^2 + 2$ with a remainder of 5, which is $x^2 + 2 + \frac{5}{x-3}$.

13 **Answer: B** 2

The x-intercepts of the parabola are given by the roots of the equation, which are $x = -2$ and $x = 6$.
The axis of symmetry is exactly halfway between the x-intercepts:
$k = \frac{-2+6}{2} = \frac{4}{2} = 2$

14 **Answer: B**

See the answer key.

15 **Answer: D** $y = 3.2(1.18)^x$

This is an exponential growth problem.
The general formula is $y = a(1 + r)^x$, where a is the initial amount, r is the growth rate, and x is the time.
Here, $a = 3.2$ million and $r = 18\% = 0.18$.
So, $y = 3.2(1 + 0.18)^x = 3.2(1.18)^x$.

16 **Answer: 9**

First, evaluate $g(-1)$:
$g(-1) = -3 - (-1) = -3 + 1 = -2$
Next, substitute this result into $f(x)$:
$f(-2) = (-2)^2 - 5(-2) - 6 = 4 + 10 - 6 = 8$

17 **Answer: 1**

Substitute $x = 0$ into the first inequality:
$4(0) + y < 2 \implies y < 2$
We also know $y \geq -2$.
The integer values for y in this range are $-2, -1, 0, 1$.
The highest integer value is 1.

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18 **Answer: 4**

Observing the given graph, the absolute maximum point of $f(x)$ is at $(-2, 4)$, so $(p, q) = (-2, 4)$.
The absolute minimum point is at $(-1, -3)$, so $(r, s) = (-1, -3)$.
We need the value of $q - p + s - r$:
$q - p + s - r = 4 - (-2) + (-3) - (-1) = 4 + 2 - 3 + 1 = 4.$

19 **Answer: 8**

First, evaluate $g(-1)$:
$g(-1) = -3 - (-1) = -3 + 1 = -2$
Next, substitute this result into $f(x)$:
$f(-2) = (-2)^2 - 5(-2) - 6 = 4 + 10 - 6 = 8$

20 **Answer: 4/5**

First, evaluate $g(-1)$:
$g(-1) = -3 - (-1) = -3 + 1 = -2$
Next, substitute this result into $f(x)$:
$f(-2) = (-2)^2 - 5(-2) - 6 = 4 + 10 - 6 = 8$

21 **Answer: C** $k = \frac{2c-6b}{a+d}$

Subtracting the triple of b from c gives: $c - 3b$.
Half of the sum of a and d is: $\frac{1}{2}(a + d)$.
Dividing the two expressions gives:
$k = \frac{c-3b}{\frac{1}{2}(a+d)} = \frac{2(c-3b)}{a+d} = \frac{2c-6b}{a+d}$

22 **Answer: A** 316

Let N be the number of new books and O be the number of old books.
$N + O = 474$
$N = 2O \implies O = \frac{N}{2}$
Substitute O in the first equation:
$N + \frac{N}{2} = 474 \implies \frac{3N}{2} = 474 \implies 3N = 948 \implies N = 316$

23 **Answer: B** 9.6π

First, complete the square to find the radius of the circle.
$(x^2 - 8x + 16) + (y^2 + 12y + 36) = 12 + 16 + 36$
$(x - 4)^2 + (y + 6)^2 = 64$
The radius squared (r^2) is 64.
The area of the entire circle is $\pi r^2 = 64\pi$.
The area of the sector is given by: $\frac{\text{central angle}}{360^\circ} \times \text{Area}$
$\frac{54}{360} \times 64\pi = \frac{3}{20} \times 64\pi = \frac{192\pi}{20} = 9.6\pi$

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24 **Answer: C** $y = -\frac{2}{5}x + 3$

Looking at the graph, the y-intercept is at $(0, 3)$. This means the equation must have $+3$ at the end, eliminating option A.
The line also passes exactly through the point $(5, 1)$.
We can verify this with the remaining options.
Using $y = -\frac{2}{5}x + 3$, substitute $x = 5$:
$y = -\frac{2}{5}(5) + 3 = -2 + 3 = 1$. This matches the point $(5, 1)$.
The slope is $m = \frac{1-3}{5-0} = -\frac{2}{5}$.

25 **Answer: D** 6

First, find the composite function $f(g(x))$:
$f(g(x)) = 5 - 2\left(\frac{x^2}{4}\right) = 5 - \frac{x^2}{2}$
Since x^2 is always greater than or equal to 0, $\frac{x^2}{2} \geq 0$.
Therefore, $5 - \frac{x^2}{2} \leq 5$.
The range of $f(g(x))$ is all real numbers less than or equal to 5, or $(-\infty, 5]$.
The value 6 is not in this range.

26 **Answer: D**

See the answer key.

27 **Answer: B** IT

History: $\frac{70}{250} = 28\%$. Math: $\frac{60}{275} \approx 21.8\%$. Audit: $\frac{30}{125} = 24\%$. IT: $\frac{40}{100} = 40\%$. IT is the highest.

28 **Answer: C** 29/65

Total preferring Rome or Paris = $150 + 175 = 325$. Audit/Math in Rome = $35 + 35 = 70$. Audit/Math in Paris = $65 + 10 = 75$. Total Audit/Math = 145. Fraction = $\frac{145}{325} = \frac{29}{65}$.

29 **Answer: D** $206,700\$$

Let P be the price and A be the area. Since they are directly proportional, $\frac{P_1}{A_1} = \frac{P_2}{A_2}$.
$\frac{168275}{127} = \frac{x}{156}$
$x = 156 \times \frac{168275}{127}$
$x = 156 \times 1325 = 206700$

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30 **Answer: C** 17

Let x be the number of additional doctors.
Initial doctors = 34
Initial nurses = 68
Initial total staff = $34 + 68 = 102$
New number of doctors = $34 + x$
New total staff = $102 + x$
The new ratio of doctors to total staff is $\frac{3}{7}$:
$\frac{34+x}{102+x} = \frac{3}{7}$
$7(34 + x) = 3(102 + x)$
$238 + 7x = 306 + 3x$
$4x = 68 \implies x = 17$
So, 17 additional doctors should join.

31 **Answer: B** $3xy^2\sqrt{3xy - 2}$

Factor the terms inside the square root by finding the greatest common factor (GCF).
The GCF of $27x^3y^5$ and $18x^2y^4$ is $9x^2y^4$.
$\sqrt{9x^2y^4(3xy - 2)}$
Take the square root of the perfect square factors ($9x^2y^4$):
$\sqrt{9x^2y^4} = 3xy^2$ (since x and y are positive)
This leaves the non-perfect square part inside the radical:
$3xy^2\sqrt{3xy - 2}$

32 **Answer: D** 30

First, evaluate $g(-1)$:
$g(-1) = -3 - (-1) = -3 + 1 = -2$
Next, substitute this result into $f(x)$:
$f(-2) = (-2)^2 - 5(-2) - 6 = 4 + 10 - 6 = 8$

33 **Answer: D** 275

The ratio is 3 parts pop to 11 parts rock, which means there are $3 + 11 = 14$ parts in total.
To find the number of songs per part, divide the total number of songs by the total number of parts:
$350 \div 14 = 25$ songs per part.
Since there are 11 parts rock songs, multiply 11 by the number of songs per part:
$11 \times 25 = 275$ rock songs.

34 **Answer: A**

See the answer key.

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35 **Answer: D** $\begin{cases} 10a + 25b \leq 250 \\ 45a + 120b \geq 900 \end{cases}$

Let a be the number of 10 kg jars and b be the number of 25 kg jars.
The total weight is $10a + 25b$, and it must not exceed 250, so: $10a + 25b \leq 250$.
The total cost is $45a + 120b$, and it must be at least $\$900$, so: $45a + 120b \geq 900$.
This corresponds to the system in option D.

36 **Answer: C** 92

Sum of the three tests = $3 \times 76 = 228$.
Let x be the fourth test grade.
$\frac{228+x}{4} = 80 \implies 228 + x = 320 \implies x = 320 - 228 = 92$.

37 **Answer: A** 11694$

Let the initial amount be x.
$x(1 + 0.225) = 14325 \implies 1.225x = 14325$
$x = \frac{14325}{1.225} \approx 11693.88$
Rounded to the nearest dollar is 11694.

38 **Answer: C** The survey is unreliable because the sample does not cover employees of different positions.

The survey wants to find the average income rate in the *entire* banking industry, but they only asked bank managers. This is a biased sample representing only the highest earners, thereby making it unreliable since it doesn't cover all different positions.

39 **Answer: D**

See the answer key.

40 **Answer: B** [30;40[

Cumulative frequency: $[20; 30[$ is 28%. $[30; 40[$ adds 37%, reaching 65%. The median (50%) falls in $[30; 40[$.

41 **Answer: B**

See the answer key.

42 **Answer: A** $f(x) = (x + 1)(x - 2)^2$

The graph touches the x-axis at $x = 2$ (double root) and crosses at $x = -1$.

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43 **Answer: B** 1 hour, 20 minutes

Total number of words = $16 \times 525 = 8400$ words.
Time in seconds = $\frac{8400}{1.75} = 4800$ seconds.
Convert to minutes: $\frac{4800}{60} = 80$ minutes.
80 minutes is 1 hour and 20 minutes.

44 **Answer: D** 4

Looking at the pdf graph, drawing a horizontal line at $y = 0.5$ intersects the depicted curve at 4 different points. Hence, there are 4 solutions.

45 **Answer: C** $f(x) = -(x+1)(x-3)$

Parabola opens downward (negative leading coefficient) with roots at $x = -1$ and $x = 3$.

46 **Answer: A** 13

First, evaluate $g(-1)$:
$g(-1) = -3 - (-1) = -3 + 1 = -2$
Next, substitute this result into $f(x)$:
$f(-2) = (-2)^2 - 5(-2) - 6 = 4 + 10 - 6 = 8$

47 **Answer: C** 16 cm

Volume of liquid in cylinder = $\pi r^2 h = \pi(4)^2(12) = 192\pi$.
The volume of the liquid in the cone will be the same.
The volume of a cone is $V = \frac{1}{3}\pi r^2 h$.
The new cone has a diameter of 12 cm, so $r = 6$ cm.
$192\pi = \frac{1}{3}\pi(6)^2 h \implies 192\pi = 12\pi h \implies h = 16$ cm.

48 **Answer: B** The subscription fees of the website

In the linear equation $f = 7g + 88$, 88 is the y-intercept, representing the initial, fixed cost before any games are downloaded. This corresponds to a fixed subscription fee for the website. 7 is the rate (cost per game).

49 **Answer: C** 6.250%

Let the original price be P.
First, increased by 25%: $P \times 1.25$.
Then, decreased by 15%: $(P \times 1.25) \times 0.85 = P \times 1.0625$.
The overall effect is multiplying by 1.0625, which corresponds to a 6.25% increase.

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50 **Answer: B** 4

From the graph, $ED = 9$ and $BC = 12$.
We know $AD = \frac{4}{3}ED = \frac{4}{3}(9) = 12$.
Since triangles ADE and ABC are similar, the ratios of corresponding sides are equal:
$\frac{AD}{AB} = \frac{ED}{BC}$
$\frac{12}{AB} = \frac{9}{12}$
$\frac{12}{AB} = \frac{3}{4} \implies AB = \frac{48}{3} = 16.$
$DB = AB - AD = 16 - 12 = 4.$

51 **Answer: 25/8**

First, evaluate $g(-1)$:
$g(-1) = -3 - (-1) = -3 + 1 = -2$
Next, substitute this result into $f(x)$:
$f(-2) = (-2)^2 - 5(-2) - 6 = 4 + 10 - 6 = 8$

52 **Answer: 35**

Let c be the number of correct answers and w be the number of incorrect answers.
$c + w = 50 \implies w = 50 - c$
$3c - 2w = 75$
Substitute w:
$3c - 2(50 - c) = 75$
$3c - 100 + 2c = 75$
$5c = 175$
$c = 35$

53 **Answer: 4**

$-3 \le 2b - 1 \le 3$
Add 1 to all parts:
$-2 \le 2b \le 4$
Divide all parts by 2:
$-1 \le b \le 2$
The integer values for b are $-1, 0, 1, 2$.
There are exactly 4 integer values.

54 **Answer: 1568**

In the sample, 80 out of 250 lawyers deal most frequently with murder.
Proportion = $\frac{80}{250} = \frac{8}{25}$.
To estimate the total out of 4900, multiply the proportion by the total population:
$\frac{8}{25} \times 4900 = 8 \times 196 = 1568.$

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55 **Answer: 7**

First, use $g(-1) = 1$ to find a:
$a(-1)^2 - 2(-1) - 5 = 1$
$a + 2 - 5 = 1$
$a - 3 = 1 \implies a = 4$
Now the function is $g(x) = 4x^2 - 2x - 5$.
Find $g(2)$:
$g(2) = 4(2)^2 - 2(2) - 5 = 4(4) - 4 - 5 = 16 - 9 = 7$.

56 **Answer: 5**

According to the Remainder Theorem, the remainder of polynomial $k(x)$ divided by $x + 2$ is equal to $k(-2)$.
$k(-2) = 3(-2)^3 + 8(-2)^2 - 2(-2) - 7$
$k(-2) = 3(-8) + 8(4) + 4 - 7$
$k(-2) = -24 + 32 + 4 - 7 = 5$

57 **Answer: 4**

First, evaluate $g(-1)$:
$g(-1) = -3 - (-1) = -3 + 1 = -2$
Next, substitute this result into $f(x)$:
$f(-2) = (-2)^2 - 5(-2) - 6 = 4 + 10 - 6 = 8$

58 **Answer: 20**

Let the longer leg be x and the shorter leg be $x - 4$.
Area = $\frac{1}{2} \times \text{base} \times \text{height} = 96$
$\frac{1}{2}x(x - 4) = 96 \implies x(x - 4) = 192$
$x^2 - 4x - 192 = 0$
$(x - 16)(x + 12) = 0 \implies x = 16$ (since length must be positive).
The legs are 16 and $16 - 4 = 12$.
The hypotenuse h is found using the Pythagorean theorem:
$h = \sqrt{16^2 + 12^2} = \sqrt{256 + 144} = \sqrt{400} = 20$.

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